Full Stack Developer

Take-Home Assignment

Estimated time: **4 to 8 hours** | Deadline: **9/4/26, 23:59**

> **A note on AI tools**
> You are welcome to use AI tools such as Claude, Copilot, or ChatGPT to help complete this assignment. We are evaluating the quality of your thinking and your decisions, not whether you used assistance.

BACKGROUND

The grain storage industry

The global grain market is worth nearly $1 trillion per year. Grains such as wheat, corn, and soybeans are produced at approximately 3.6 billion tonnes annually and stored in large warehouses for months to over a year before reaching food centers, dairy farms, and other end users.

When grain is stored in massive quantities, the pressure and volume generate internal heat. Combined with moisture, this creates conditions for mold growth and in extreme cases, spontaneous combustion. Continuous monitoring from inside the grain mass is therefore critical.

How agriQ works

agriQ's approach is different from other solutions on the market. Instead of measuring only from the surface or through fixed cables, agriQ uses small wireless sensor balls that are thrown directly into the grain pile and settle at different depths. This gives real visibility into the core of the pile, where heat and moisture problems actually start, not just the outer layer. The technology behind this approach is patent-protected.

Each pile has sensor balls distributed across three layers: bottom, middle, and top. The system also collects ambient readings from gateway sensors mounted inside the storage cell, outside the grain pile, and pulls in external data from weather APIs and commodity markets.

Thresholds

Temperature (degrees Celsius)

Range	Status	Meaning
Below 30	**OK**	Grain is stable.
30 to 45	**Warning**	Conditions deteriorating, increase monitoring.
Above 45	**Critical**	Risk of fire or total spoilage, act immediately.
Goal		The system must warn before reaching 45 so operators can act in time.

Moisture (%)

Range	Status	Meaning
Below 14%	**OK**	Grain is dry and stable, safe for long-term storage.
14% to 17%	**Warning**	Mold growth becomes possible, increase monitoring.
Above 17%	**Critical**	High spoilage risk, immediate action required.
Note		High temperature and high moisture together are far more dangerous than either alone. Your risk logic must reflect this.

TASK 1 | BACKEND ARCHITECTURE DESIGN

No code required. Write a design document of one to two pages in any format you prefer, such as a written document, a diagram with notes, or a combination of both.

Facility context

```
Address:    Harish 7, Emek Hefer Industrial Park, Israel
Piles:      4 wheat piles
Cell size:  50m long  x  25m wide  x  10m high  (approx. 3,000 tonnes per pile)
Sensors:    30 sensor balls per pile
            S01 to S10   bottom layer
            S11 to S20   middle layer
            S21 to S30   top layer
Frequency:  One reading every 12 hours
```

Data sources

Your system must consolidate the following data sources into a single platform:

- Sensor ball readings from inside each grain pile, meaning temperature and moisture per sensor ball
- Gateway readings, meaning ambient temperature and humidity inside the cell but outside the pile
- Weather API data for the Emek Hefer area
- Commodity price data from the Chicago Board of Trade (CBOT)
- Any other source you believe is relevant, with a short justification

System architecture

Describe the architecture you would build for this system. Walk us through your thinking as if explaining it to a colleague: what pieces would you create, how would data move from a sensor ball reading all the way to an alert on an operator's screen, and what would the database structure look like. We want to see how you think about the problem end to end, including how you would handle sensor data ingestion reliably, and where future integrations with ERP systems and commodity markets would fit into the picture. If there are trade-offs or decisions you would need to make, describe them and explain your reasoning.

Risk logic

Describe how you would build the logic that decides whether a grain pile is at risk. Think about what it means for a pile to be in danger: it is not just one sensor reading one high value at one moment in time. Explain how you would combine readings from multiple sensor balls to form a picture of the pile's condition. Consider how you would tell the difference between a sensor ball that is malfunctioning and a real problem developing inside the grain. Think about what patterns in the data over time would indicate increasing risk, and how you would decide when to trigger an alert versus when to continue observing. Describe how your logic would handle a situation where conditions are worsening gradually versus one where something changes quickly.

TASK 2 | OPERATOR DASHBOARD

Design principle
The people using this system are grain storage operators, not software users. The dashboard should be intuitive and easy to use for someone who is not technical. Keep that person in mind when making every design decision.

Build a frontend dashboard using the mock data provided in this document. No backend is required. **The application must be built in React 19 with Node.js.**

Two pages required

Page 1	**Sites** Shows all storage sites and piles. For each pile, the current status and readings must be visible. When a pile is selected, show the sensor balls and their current readings inside that pile. The storage cell measures 50m long by 25m wide by 10m high, with 30 sensor balls across three layers. How you choose to present the sensor positions is up to you.
Page 2	**Alerts** Shows all active alerts across all piles. Each alert must clearly state which pile is affected, which sensors are involved, what the current reading is, and what the operator should do next. Design this page for a person who walks in and needs to act immediately.

Mock data

The table below contains the current readings for all four piles. Use these values as your hardcoded data in the application.

Pile	Status	Most sensors	Problem sensors
Emek North	**OK**	21 C / 12.5% moisture	None
Emek South	**Warning**	28 C / 13.2% moisture	S01 to S04 (bottom): 44 C / 16.1% moisture
Emek East	**Critical**	26 C / 13.0% moisture	S11 to S15 (middle): 51 C / 18.4% moisture S28 (top): erratic readings, possible faulty sensor
Emek West	**Warning**	35 C / 14.8% moisture	S06 to S08 (bottom): 39 C / 16.2% moisture

SUBMISSION

Push your work to a public GitHub repository and send us the link to noapeleg@agriq.farm

Task 1	Any readable format: a written document, a Google Doc link, a PDF, or a diagram with notes.
Task 2	Include a clear explanation of how to run the application locally, including all required setup steps.

What we evaluate
Clarity of thinking, quality of decisions, and design for non-technical users. We are not looking for perfection.

For any Questions, Please Email us noapeleg@agriq.farm